

Option value = b

Evergreen

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American Put Option Pricing

Given:- $S_0 = ₹100$, $K = ₹100$, $r = 6\%$, $\sigma = 25\%$ per year

$T = 1$ year, $N = 3$ steps

$$\Delta t = \frac{T}{N} = \frac{1}{3}$$

$$u = e^{r\sqrt{\Delta t}} = e^{\frac{1}{4}\sqrt{\frac{1}{3}}} = 1.1553$$

$$d = \frac{1}{u} = 0.8656$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.06 \times \frac{1}{3}} - 0.8656}{1.1553 - 0.8656} = 0.5337$$

$$\text{disc} = e^{-r\Delta t} = e^{-0.02} = 0.9802$$

Recombination :-

$$\text{As } u \times d = 1.1553 \times 0.8656 = 1$$

The tree recombines.

European Pricing :-

By Backward Induction.

At $t = N$

$$f \text{ value} = \max(K - S, 0)$$

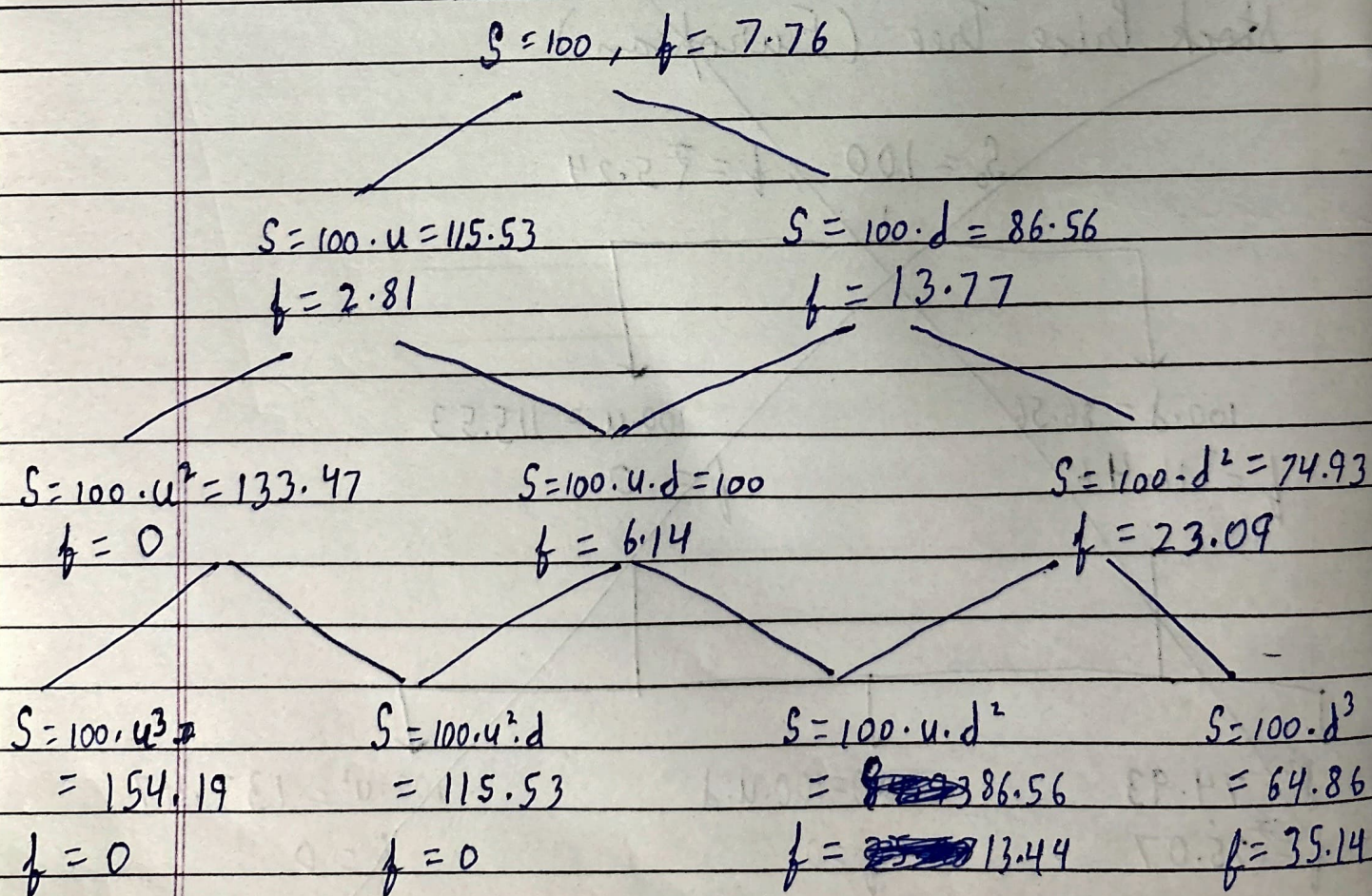
$$\text{e.g. for up-up-up, value} = \max(-33.47, 0) = 0$$

At $t = N - 1$

$$\text{value} = e^{-rt} (p \cdot f_{\text{up},N} + (1-p) \cdot f_{\text{down},N})$$

and so on

$$\text{e.g. for up-up, put value} = 0.9802(0.53370 + 0) = 0$$



American Pricing :-

$$\text{Value} = \max(\text{Exercise Value}, \text{Continuation Value})$$

Here, Exercise Value, $E = \max(K - S, 0)$

$$\text{Continuation Value, } C = e^{-r_0 t} (p \times b_{\text{up}} + (1-p) b_{\text{down}})$$

For $t = N$

European put price = American price.

For $t = N-2 = 1$

e.g. for $S = 100$ i.e. "up"

$$\text{value} = \max(0, 0.9802 (0.5337 \times 0 + 0.4663 \times 6.14))$$

$$= ₹ 2.81$$

$S=100, E=0, K=8.16,$

Hold

$S=115.53$

$E=0, K=2.81$

Hold

$S=86.56$

$E=13.44, K=14.67$

Hold

$S=133.47$
 $E=0, K=0$

Exercise/Hold

$S=100$

$E=0, K=6.14$

Hold

$S=74.93$

$E=25.07, K=23.09$

Exercise

$S=154.19$

$E=0, K=0$

Exercise/Hold

$S=115.53$

$E=0, K=0$

Exercise/Hold

$S=86.56$

$E=0, K=13.44$

Exercise

$S=64.86$

$K=0, E=35.14$

Exercise

Early Exercise Premium :-

= American Price - European Price

$$= 8.16 - 7.76 = \text{₹} 0.40$$

Greeks :- Delta

$$\Delta = \frac{f_u - f_d}{S_u - S_d}$$

Here, f_u & f_d = up and down node values of option at that step.

S_u & S_d = up & down node values of stock.

$$\Delta = \frac{2.81 - 14.67}{115.53 - 86.56} = -0.4094$$

$\Delta < 0$ as it should for puts.

Early exercise became optimal when the option was significantly in the money and very close to expiry. In this case, once the stock price had fallen well below the strike price near the final stage of the tree, exercising immediately provided greater value than holding the option further. This reinforces the intuition gained earlier in the course that American put options are the only options where early exercise may be beneficial when the put is deep ITM.

I am extremely sorry for being late two times in a row. I was preoccupied with a finance competition held by finance club IITB.